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3. (M, d) is a connected metric space and has at-least two elements. Show that M is uncountable. [3]

(M,d) has at least two elements, ~~i.e $a, b \in M$ & $a \neq b$.~~ ~~a_1, b_1~~

i.e $a_1, a_2 \in A$ & $a_1 \neq a_2$

Proof by contra positive:

To show if M is atmost countable then M is disconnected.

let M be atmost countable.

i.e. ~~$M = \{a, b, c, \forall i$~~

$M = \{a_i : i \in \mathbb{N}\}$

let $A = \{a_1\}$

$B = \{a_2, a_3, \ldots\}$

$A \cap B = \phi$

and $A \cup B = M$

$\Rightarrow M$ is disconnected.

(this includes case of finite but ~~let me et~~ more elaboration below)

Let M be finite.

$M = \{a_i : 1 \leq i \leq n\}$ $n \geq 2$

let $A = \{a_1\}$

$B = \{a_2, \ldots a_n\}$

again $A \cap B \neq \phi$

$A \cup B = M$

$\Rightarrow M$ is disconnected.

4. Let (M, d) be a metric space and $A \subseteq M$. Prove that if A is totally bounded then (the closure) $\overline{A}$ is totally bounded. [3]

A is totally bounded.

~~$\Rightarrow \exists n$~~ $\forall \varepsilon > 0$,

$\exists$ ~~n, n_2~~ $x_1, x_2, \ldots x_n$

s.t. $A \subseteq \bigcup_{i=1}^{n} B_\varepsilon(x_i)$

to show $\forall \varepsilon > 0$

$\exists y_1, y_2, \ldots y_m$

s.t $\overline{A} \subseteq \bigcup_{i=1}^{m} B_\varepsilon(y_i)$

~~take $y \in$~~

~~ε~~ $\varepsilon > 0$

for $x \in \overline{A}$ $\left(A \subseteq \bigcup_{i=1}^{n} B_{\varepsilon/2}(x_i)\right)$

$B_{\varepsilon/2}(x) \cap A \neq \phi$

$B_{\varepsilon/2}(x) \cap \bigcup_{i=1}^{n} B_{\varepsilon/2}(x_i) \neq \phi$

$\Rightarrow B_{\varepsilon/2}(x) \cap B_{\varepsilon/2}(x_i) \neq \phi$

for some $1 \leq i \leq n$

~~take~~. $\exists y$ s.t

$y \in B_{\varepsilon/2}(x)$ &

$y \in B_{\varepsilon/2}(x_i)$

~~$d(x, y)$~~

$d(x, x_i) \leq d(x, y) + d(x_i, y)$

$< \varepsilon$

$\Rightarrow d(x, x_i) < \varepsilon$

$\Rightarrow x \in B_\varepsilon(x_i)$ for some $1 \leq i \leq n$

$\Rightarrow \overline{A} \subseteq \bigcup_{i=1}^{n} B_\varepsilon(x_i)$

$\Rightarrow \overline{A}$ is totally bounded.

5. Let (M, d) be a complete metric space. A subset $A \subseteq M$ is complete if and only if A is a closed set.

[4]

M is a complete metric space.

$\Leftrightarrow$ any cauchy sequence (x_n) in M converges to $x \in M$.

A subset $A \subseteq M$ is complete. if any cauchy sequence (x_n) in A converges to a point ~~A~~. in A.

let A be complete.

take. any convergent sequence $\Leftrightarrow$ cauchy sequence

and we know if ~~all~~ convergent sequence x_n ~~$\to x$~~ &

$(x_n) \subset A$ with $x_n \to x$

and $x \in A$ then

A is a closed set

(A is a closed set $\Leftrightarrow$ if $(x_n) \subset A$ & $x_n \to x$ then $x \in A$?)

A is a closed set.

and (M, d) is a complete metric space.

~~consid~~ ~~any~~ consider any cauchy sequence (x_n) in A.

~~$\Rightarrow$~~ we know $(x_n) \to x$ for some $x \in M$

~~but~~ as (M, d) is complete.

but as A is closed.

$x \in A$ ~~(all limit points~~ of ~~sequence in~~ A

$\Rightarrow$ A is complete.

········· Rough Work ·········

A is totally bounded

$\exists\, x_1, x_2, x_3, \ldots x_n \in M$ s.t $\bigcup B_\varepsilon(x_i) \supseteq A$.

$\exists\, y_1, y_2, \ldots y_n \in M$ s.t $\bigcup B_\varepsilon(y_i) \supseteq \bar{A}$

$x \in \bar{A} \Rightarrow B_\varepsilon(x) \cap A \neq \phi \;\; \forall \varepsilon > 0$.

$d(x, a) < \varepsilon$

$\exists\, a$

$B_\varepsilon(x) \cap \left(\bigcup B_\varepsilon(x_i)\right) \neq \phi$

$y \in B_\varepsilon(x)$ s.t $y \in B_\varepsilon(x_i)$

$x \in B_\varepsilon(y)$